

Class 14: Pathway Analysis from RNA-Seq Results

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Overview

Pathway analysis aims to reduce the complexity of interpreting large gene lists by mapping them to known biological pathways and processes. Here we use DESeq2 for differential expression, then GAGE for KEGG pathway enrichment, and finally explore Gene Ontology (GO) and Reactome for further biological insight.

Section 1. Differential Expression Analysis

```
library(DESeq2)
library(ggplot2)
```

Load the data

```
# row.names=1 sets the first column as row names instead of a data column
colData <- read.csv("GSE37704_metadata.csv", row.names = 1)
countData <- read.csv("GSE37704_featurecounts.csv", row.names = 1)

head(colData)
```

```

condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369 hoxa1_kd
SRR493370 hoxa1_kd
SRR493371 hoxa1_kd

```

```
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

Q1. Remove the \$length column

The `length` column is transcript length info — not a sample. We remove it so `countData` columns match `colData` rows exactly, as DESeq2 requires.

```

# -1 removes the first column (length)
countData <- as.matrix(countData[, -1])
head(countData)

```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Q2. Filter zero-count genes

Genes with zero counts across all samples can't be tested and hurt our multiple-testing correction, so we remove them upfront.

```
# rowSums > 0 keeps only genes detected in at least one sample
countData <- countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

Run DESeq2

```
# Always check files line up before running - mismatched order = wrong results
all(colnames(countData) == rownames(colData))
```

```
[1] TRUE
```

```
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData   = colData,
                              design    = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds <- DESeq(dds)
```

Get results

```
# Positive log2FoldChange = higher in hoxa1_kd vs control_sirna
res <- results(dds)
summary(res)
```

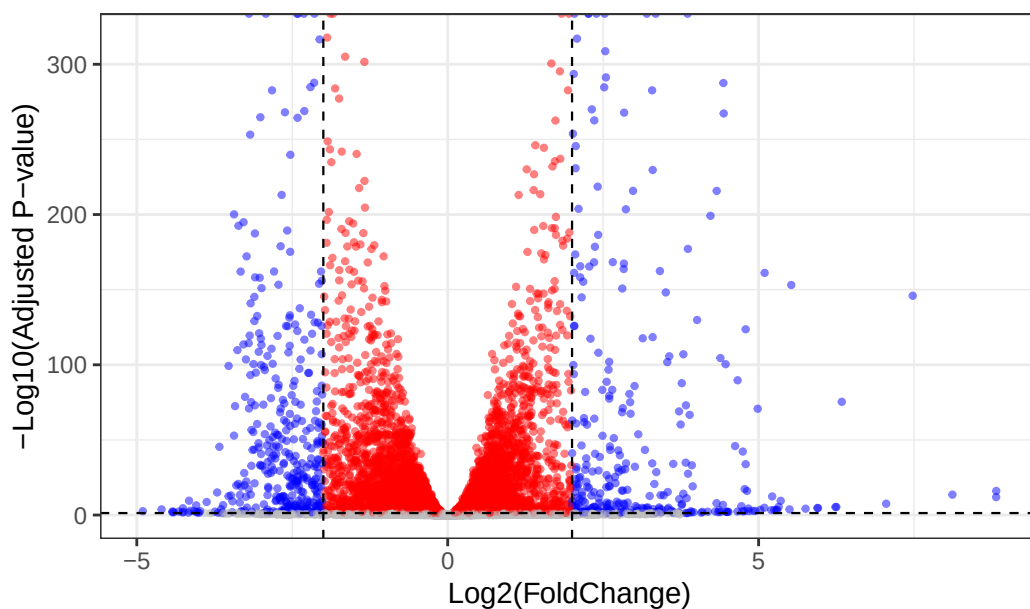
```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

```
# Color scheme: gray = not significant, red = padj < 0.05,
# blue = padj < 0.05 AND |log2FC| > 2 (significant + large effect)
mycols <- rep("gray", nrow(res))
mycols[!is.na(res$padj) & res$padj < 0.05] <- "red"
mycols[!is.na(res$padj) & res$padj < 0.05 & abs(res$log2FoldChange) > 2] <- "blue"

ggplot(as.data.frame(res)) +
  aes(x = log2FoldChange,
      y = -log10(padj)) +
  geom_point(color = mycols, alpha = 0.5, size = 0.8) +
  xlab("Log2(FoldChange)") +
  ylab("-Log10(Adjusted P-value)") +
  ggtitle("Volcano Plot: HOXA1 Knockdown vs Control") +
  geom_vline(xintercept = c(-2, 2), linetype = "dashed", linewidth = 0.4) +
  geom_hline(yintercept = -log10(0.05), linetype = "dashed", linewidth = 0.4) +
  theme_bw()
```

Volcano Plot: HOXA1 Knockdown vs Control



Add gene annotations

```
# We need SYMBOL for readability, ENTREZID for KEGG, GENENAME for context
library(AnnotationDbi)
library(org.Hs.eg.db)

res$symbol <- mapIds(org.Hs.eg.db,
                     keys       = row.names(res),
                     keytype    = "ENSEMBL",
                     column     = "SYMBOL",
                     multiVals  = "first")

res$entrez <- mapIds(org.Hs.eg.db,
                     keys       = row.names(res),
                     keytype    = "ENSEMBL",
                     column     = "ENTREZID",
                     multiVals  = "first")

res$name    <- mapIds(org.Hs.eg.db,
                     keys       = row.names(res),
                     keytype    = "ENSEMBL",
```

```

      column = "GENENAME",
      multiVals = "first")

head(res, 10)

```

log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51281e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02
ENSG00000188157	9128.439422	0.3899088	0.0467164	8.346302	7.04333e-17
ENSG00000237330	0.158192	0.7859552	4.0804729	0.192614	8.47261e-01
	padj	symbol	entrez	name	
	<numeric>	<character>	<character>	<character>	
ENSG00000279457	6.86555e-01	NA	NA	NA	
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..	
ENSG00000188976	1.76553e-35	NOC2L	26155	NOC2 like nucleolar ..	
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..	
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..	
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..	
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..	
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..	
ENSG00000188157	4.21970e-16	AGRN	375790	agrin	
ENSG00000237330	NA	RNF223	401934	ring finger protein ..	

Save results

```

res <- res[order(res$pvalue), ]
write.csv(res, file = "deseq_results.csv")

```

Section 2. Pathway Analysis

GAGE tests whether genes in a known pathway change together in a coordinated direction — more powerful than testing individual genes.

```
library(pathview)
library(gage)
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]

# GAGE needs a named vector: values = fold changes, names = Entrez IDs
foldchanges      <- res$log2FoldChange
names(foldchanges) <- res$entrez

# same.dir = TRUE tests for pathways moving consistently up OR down
keggres <- gage(foldchanges, gsets = kegg.sets.hs)
```

```
head(keggres$less)      # top down-regulated pathways
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

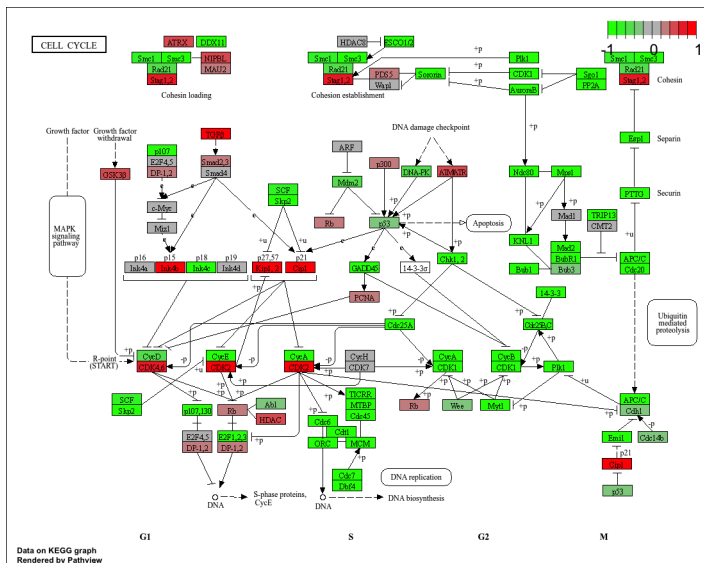
```
head(keggres$greater) # top up-regulated pathways
```

		p.geomean	stat.mean	p.val
hsa04640	Hematopoietic cell lineage	0.002822776	2.833362	0.002822776
hsa04630	Jak-STAT signaling pathway	0.005202070	2.585673	0.005202070
hsa00140	Steroid hormone biosynthesis	0.007255099	2.526744	0.007255099
hsa04142	Lysosome	0.010107392	2.338364	0.010107392
hsa04330	Notch signaling pathway	0.018747253	2.111725	0.018747253
hsa04916	Melanogenesis	0.019399766	2.081927	0.019399766
		q.val	set.size	exp1
hsa04640	Hematopoietic cell lineage	0.3893570	55	0.002822776
hsa04630	Jak-STAT signaling pathway	0.3893570	109	0.005202070
hsa00140	Steroid hormone biosynthesis	0.3893570	31	0.007255099
hsa04142	Lysosome	0.4068225	118	0.010107392
hsa04330	Notch signaling pathway	0.4391731	46	0.018747253
hsa04916	Melanogenesis	0.4391731	90	0.019399766

Pathview — Cell Cycle pathway

```
# Downloads the KEGG pathway diagram and colors genes by fold change
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

```
knitr::include_graphics("hsa04110.pathview.png")
```



Top 5 up-regulated pathways

```
keggrespathways_up <- rownames(keggres$greater)[1:5]
keggresids_up      <- substr(keggrespathways_up, start = 1, stop = 8)
keggresids_up
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids_up, species = "hsa")
```

Q. Top 5 down-regulated pathways

```
# Same approach as above but from keggres$less
keggrespathways_dn <- rownames(keggres$less)[1:5]
keggresids_dn      <- substr(keggrespathways_dn, start = 1, stop = 8)
keggresids_dn
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa03440" "hsa04114"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids_dn, species = "hsa")
```

Section 3. Gene Ontology (GO)

GO annotates genes with Biological Process (BP), Cellular Component (CC), and Molecular Function (MF) terms. Here we focus on BP, which gives finer resolution than KEGG (thousands of terms vs ~200 pathways).

```
data(go.sets.hs)
data(go.subs.hs)

gobpsets <- go.sets.hs[go.subs.hs$BP]
gobpres  <- gage(foldchanges, gsets = gobpsets)

lapply(gobpres, head)
```

\$greater

		p.geomean	stat.mean	p.val
G0:0007156	homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009	morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729	tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610	behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562	epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295	tube development	5.953254e-04	3.253665	5.953254e-04
		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009	morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729	tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610	behavior	0.1967577	426	1.925222e-04
G0:0060562	epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295	tube development	0.3565320	391	5.953254e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Section 4. Reactome Analysis

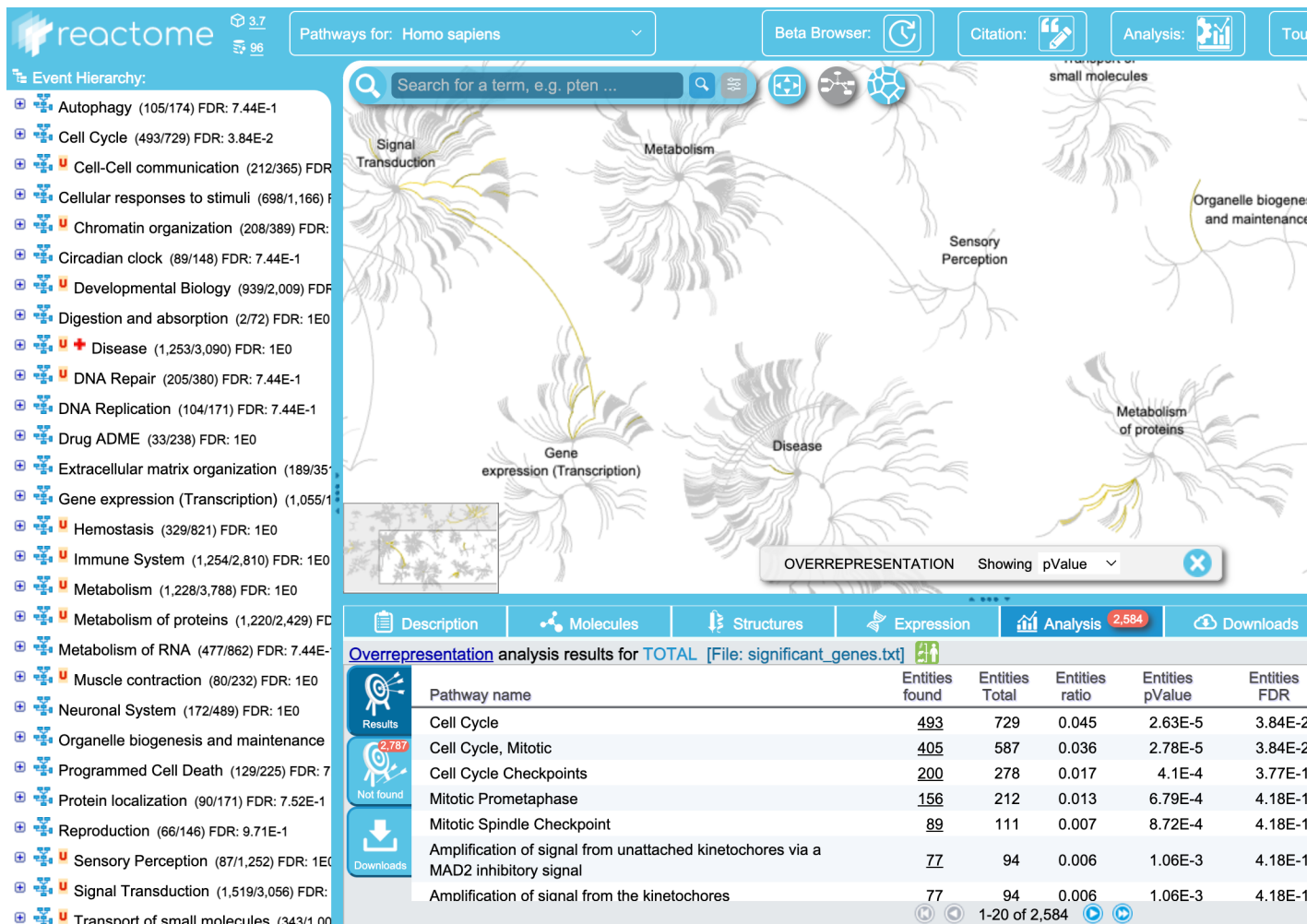
```
# Export significant genes for upload to Reactome website
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes,
            file       = "significant_genes.txt",
            row.names = FALSE,
            col.names = FALSE,
            quote      = FALSE)
```

Upload `significant_genes.txt` to <https://reactome.org/PathwayBrowser/#TOOL=AT>, select “Project to Humans” and click Analyze.

```
knitr::include_graphics("reactome_screenshot.png")
```



Q. What pathway has the most significant Entities p-value? Do results match KEGG? What factors could cause differences?

The most significant pathway is **Cell Cycle / Mitotic Cell Cycle**, consistent with our KEGG results. Both methods converge on cell division disruption as the main consequence of HOXA1 loss.

Differences between methods can arise from:

1. **Different gene set curation** — pathway membership is defined independently in each database
2. **Different statistical methods** — GAGE uses fold changes across all genes; Reactome uses a hypergeometric test on the significant gene list only

3. **Different ID mapping** — small differences in Ensembl to symbol conversion can include or exclude genes
 4. **Pathway hierarchy** — Reactome is more granular, splitting one KEGG pathway into many sub-pathways
 5. **Database update frequency** — each database is updated on its own schedule, so gene membership can differ
-

Session Information

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Los_Angeles
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats4      stats      graphics  grDevices  utils      datasets  methods
```

```
[8] base
```

```
other attached packages:
```

```
[1] gageData_2.48.0      gage_2.60.0
[3] pathview_1.50.0      org.Hs.eg.db_3.22.0
[5] AnnotationDbi_1.72.0 ggplot2_4.0.2
[7] DESeq2_1.50.2        SummarizedExperiment_1.40.0
[9] Biobase_2.70.0       MatrixGenerics_1.22.0
[11] matrixStats_1.5.0    GenomicRanges_1.62.1
[13] Seqinfo_1.0.0        IRanges_2.44.0
[15] S4Vectors_0.48.1     BiocGenerics_0.56.0
```

[17] generics_0.1.4

loaded via a namespace (and not attached):

[1] KEGGREST_1.50.0	gtable_0.3.6	xfun_0.57
[4] lattice_0.22-9	bitops_1.0-9	vctrs_0.7.2
[7] tools_4.5.3	parallel_4.5.3	tibble_3.3.1
[10] RSQLite_3.52.0	blob_1.3.0	pkgconfig_2.0.3
[13] Matrix_1.7-4	RColorBrewer_1.1-3	S7_0.2.1
[16] graph_1.88.1	lifecycle_1.0.5	compiler_4.5.3
[19] farver_2.1.2	Biostrings_2.78.0	codetools_0.2-20
[22] htmltools_0.5.9	RCurl_1.98-1.18	yaml_2.3.12
[25] GO.db_3.22.0	crayon_1.5.3	pillar_1.11.1
[28] BiocParallel_1.44.0	DelayedArray_0.36.1	cachem_1.1.0
[31] abind_1.4-8	tidyselect_1.2.1	locfit_1.5-9.12
[34] digest_0.6.39	dplyr_1.2.1	labeling_0.4.3
[37] fastmap_1.2.0	grid_4.5.3	cli_3.6.5
[40] SparseArray_1.10.10	magrittr_2.0.5	S4Arrays_1.10.1
[43] XML_3.99-0.23	withr_3.0.2	scales_1.4.0
[46] bit64_4.6.0-1	rmarkdown_2.31	XVector_0.50.0
[49] httr_1.4.8	bit_4.6.0	otel_0.2.0
[52] png_0.1-9	memoise_2.0.1	evaluate_1.0.5
[55] knitr_1.51	rlang_1.1.7	Rcpp_1.1.1
[58] glue_1.8.0	DBI_1.3.0	Rgraphviz_2.54.0
[61] KEGGgraph_1.70.0	rstudioapi_0.18.0	jsonlite_2.0.0
[64] R6_2.6.1		